SR Eyetracker

About

Eye Trackers: Manufacturer: SR Research. Non-ferrous Infra-red eye tracking system; monocular/binocular, up to 2000Hz sample rate (monocular), linked to Matlab through PsychToolbox, available as analogue output to Spike software, ideal for fixation and full screen eye tracking, pupilometry and as a physiological monitor for all volunteers.

Eyetracking data is saved into an edf file on the eyetracker PC and copied onto the Stimulus PC by a suitable command in your script (see below.) The edf file is a proprietary (SR Research) format. To extract the data into an ASCII text file use the EDF2ASC program installed on the Stimulus PC start menu, in the SR Research folder.

User Guides

Manufacturer Resources

Users of the eye track systems are expected to understand how to configure and operate both hardware and software.

The eyelink user manual provides information on the available recording options and software operation and the SR Research resource webpage provides further manuals and videos for eyetracking operation and eyelink softwares.

User Guide - 3T MRI

follow these instructions

- The system is configured with an eye to screen distance of 550mm.
- The eye track camera and illuminator is positioned external to the MRI bore with a direct line of sight to the coil mirror.
- Eye tracking should only be carried out with a head coil mirror labelled for “eye tracking”. (The standard Siemens head coil mirror may appear to work but eye tracking performance will be degraded and unreliable – see tech note below.)

Before scanning:

1. Remove the Siemens mirror and replace with the mirror labelled “eye tracking”
2. Adjust the position of the mirror so it is centred over the eyes

3. Move bed to the isocentre
4. Review eye tracking image and adjust focus ring if necessary
    1. Open 'Track'software
    2. Display camera visual to projected image
    3. Carefully adjust the camera focus ring (do not move the camera assembly)
5. Run the eye tracker calibration as normal using the 9 or 5-point calibration screen

After scanning:

1. Remove eye tracking mirror
2. Replace the standard Siemens mirror

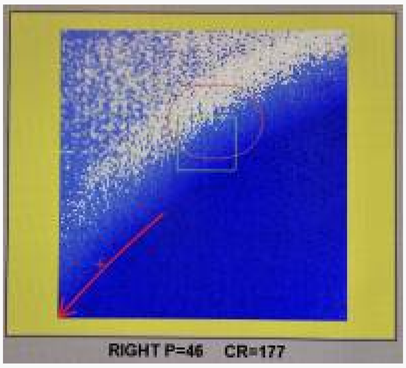
User Guide - 7T MRI

follow these instructions

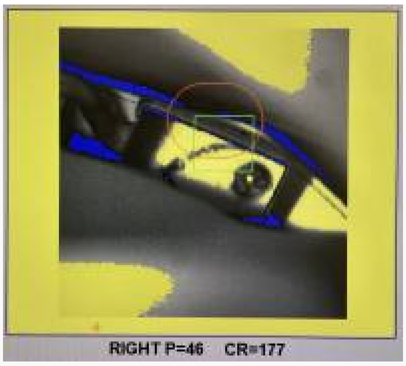
- The system is configured with an eye to screen distance of 1130mm.
- The eye tracker is mounted on a periscope frame. The periscope and attached eye tracker are positioned within the rear of the bore. The camera has a direct line of sight to the coil mirror. The illuminator targets the eye through the two periscope mirrors.
- Eye tracking should only be carried out with a head coil mirror labelled for “eye tracking”. (The standard Siemens head coil mirror may appear to work but eye tracking performance will be degraded and unreliable – see tech note below.)
- Success of eye tracking in the very tight confines of the 7T head coil is very dependent on the subject’s head shape and size. It is vital to position the head as far into the coil as possible (most superior position.)

Before scanning:

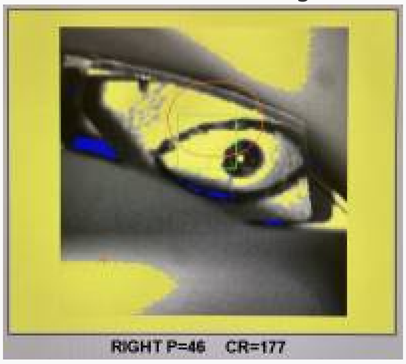
1. Turn the eyetracker PC on and check that the camera is looking at the side of the end of the bore. The (blurred) edge of the bore should be aligned with the lower left corner of the camera image.



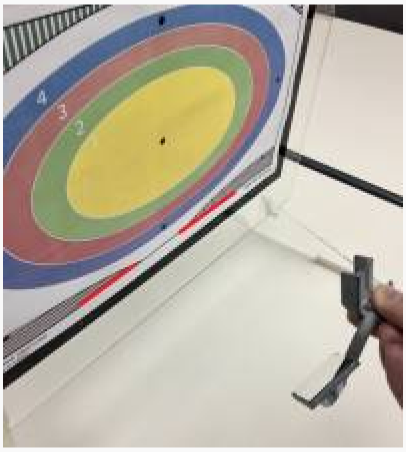
2. Remove Siemens mirror and place in the labelled “7T Coil Mirror” box in the magnet room
3. Place the “dummy screen” into the scanner bore touching the back of the bed
4. Fit either the “TMF” mirror or the “SR” mirror (located in the “7T Coil Mirror” box)
    1. The “SR” mirror will accommodate a greater range of head shapes/sizes



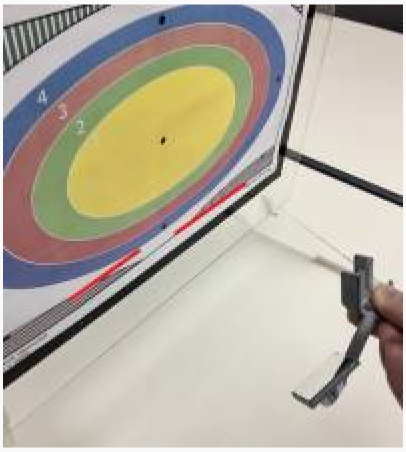
   2. The “TMF” mirror will not suffer from mirror misalignment



5. If using the “SR” mirror check the alignment of the two mirrors before fitting the mirror to the head coil
   1. OK to use



   2. Do not use



6. If the mirrors are not aligned correctly then use the “TMF” mirror or do not eye track.
7. Ask the subject to lie on the bed, and position their head as far into the head coil as possible
8. Adjust the position of the mirror so it is centred over the eyes
9. Ask the subject how much of the dummy screen can be seen, paying particular attention to

the 9 eye tracker calibration dots
10. Readjust head and/or mirror position to maximize visible area
11. Remove dummy screen
12. Move bed to the isocentre
13. Check that the subject’s eye is visible in the camera image (see right)
14. Run the eye tracker calibration as normal using the 5-point calibration screen

After scanning:

1. Remove eye tracking mirror and store in the labelled box
2. Replace the standard Siemens mirror

User Guide - Testing Rooms

follow these instructions

- An eye tracking system is available for use in Testing Room 2 and the EEG room.
- The eye tracker is connected and ready for use but must be configured and set-up by the user to meet their experimental specification.
- Due to the variability in room usage, the user is expected to remove the eye track set-up at the end of each booking period.

Experimental Control of the Eyetracker

The eyetracker can be started, stopped, and receive event marks or triggers in various way.

Psychtoolbox script control

Psychtoolbox can control the eyetracker through its eyelink command. This allows for a connection to be made to the eyetracker, for a filename to be chosen, the file to be opened, recording started/stopped, triggers sent, the file to be closed, and copied to the Stim PC for conversion from EDF format to ASCII text. These commands are typically written within a testing paradigm script.

Examples of Psychtoolbox script control can be seen on the as track2.m and track3.m scripts on the WHN Github.

Manual control

The eyetracker can be controlled entirely though its own user interface. This includes choosing a filename, calibration, starting recording (manually in time with the scan), stopping recording, and closing/saving the file. If you wish to record scanner triggers/event marks into the eye tracking file, then further steps will be required, see “Event marking” below.

Dual script control

If you are using a stimulus package other than Psychtoolbox, script control of the eye tracker may not be available e.g. Unit. One solution is to use two scripts/software in parallel to achieve eye tracker control. For example, a Unity package presents a stimuli and a simple Psychtoolbox script controls the eyetracker.

This method introduces more automation in eyetracker control, and reduces the chance of losing data through user error when compared to manual control i.e forgetting to start, stop, or save on the eyetracker PC. It is a researcher responsibility to ensure PC performance and stimuli presentation is adequate when running dual script.

User Guide - PsychToolbox

The eyetracker PC can be controlled manually on the PC itself, or from a PsychToolbox script on the Stim PC. There are examples of the resulting eyetracker files (after conversion to ascii) and corresponding example scripts are here.

Note that as the Eyetracker PC runs the DOS or Linux, not Windows, operating system the filename you give use in the Eyelink('Openfile', 'test') command MUST be 8 characters or less. If you choose a longer filename no file will be created and no eyetracking data will be saved.

NB It is vitally important to delete the file on the eyetracker PC after it has been copied to the Stim PC! The eyetracker saves these files in the data directory. e.g. at the DOS command prompt on the eyetracker PC type:

cd ..\data del test.edf cd ..\exe This will change directory to the data directory, delete the file [test.edf] created by the short script above, and change directory back to the eyetracker app directory. You can re-start the eyetracking software by typing elcl. If you have used different file names you will need to modify the command accordingly. Please ask if you have any problems.

Some people prefer to start and stop the recording for each trial, or block, rather than one long recording run as above.

Note that the MEG system also records the analogue point-of-gaze outputs from the SR eyetracker as extra data channels. Recorded as; ADC001= X-Axis, ADC002=Y-Axis & ADC003=Pupil diameter/area. Please ensure these channels are active in your set-up protocol and connected prior to recording.

Event Marking

The Eye Link recording can be time locked to stimuli via MATLAB i.e Eyelink ('StartRecording').
If using a non-MATLAB package (i.e Unity) it is possible to mark events on the Eye Link recording

montage (edf file).

MRI

If using 'Keypress triggers' for either 'Slice' or 'Volume'; the same trigger event can be sent to the Eye Link PC. By default this feature is not enabled.

The Eye Link recording montage will show EVENT and BUTTON at times that correlate to the keypress trigger sent by MRI.

If event marking is required:

1. Connect the BNC connector labelled 'Eye Link Event' to the 'Keypress Trigger Unit'
2. Ensure the options to record 'BUTTON' and 'EVENT' are selected.
3. On the bottom right corner of the recording screen, a '5' will flash when an EVENT/BUTTON is received.
4. After scanning please **DISCONNECT** the BNC cable from the Eyelink PC.

After converting the resulting EDF file to ASCII (using the edf2asc app) the text file will contain BUTTON events e.g.

```
BUTTON  77788   5       1
BUTTON  77792   5       0
```

This shows a scanner trigger at 77788 ms (button 5 being pressed). Ignore the button release (… … 5 0) a few ms later

MEG

Event marking the eye link file has not previously been required as the MEG system records the analogue point-of-gaze outputs from the SR eyetracker as extra data channels. Recorded as; ADC001= X-Axis, ADC002=Y-Axis & ADC003=Pupil diameter/area. Please ensure these channels are active in your set-up protocol and connected prior to recording.

Eye tracking data (.edf) files

Eyetracking data is saved as an .edf file. This is a proprietary format of the SR Research systems.
The .edf files are saved on the eyetracker PC and copied onto the Stimulus PC by a suitable 'ReceiveFile' command in your script.
To extract the data into an ASCII text file use the EDF2ASC program installed on the Stimulus PC start menu.
The EDF2ASC program can found via the SR Research directory, or via a Windows search.

How to free space on the eyetracker PC

Research fellows should be deleting their own edf files from the eyetracker PCs.

If the SR eyetracker partition is full:

1. Reboot the SR eyetracker PC and select "Windows" from the boot menu
2. Open Windows Explorer
3. Move the EDF files from D:\elcl\data (or E:\elcl\data) to the C: drive in Documents\Eyelink\elcl\data
4. From the Windows start menu, reboot and select "Eyetracker" from the boot menu

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Dummy Screen for 7T MRI

The transmit and receive head coils used on the 7T system impose some restriction on the visual field.
It is important that volunteers are positioned in the head coil so that they will be able to see the projected image.

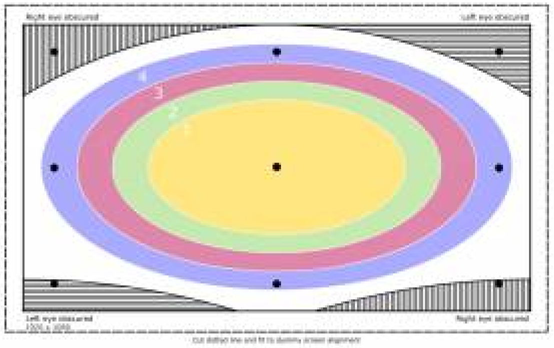
To aid volunteer positioning, a 'dummy screen' has been created.
The dummy screen is a replica of the projected image that will be visible in-bore. It shows:

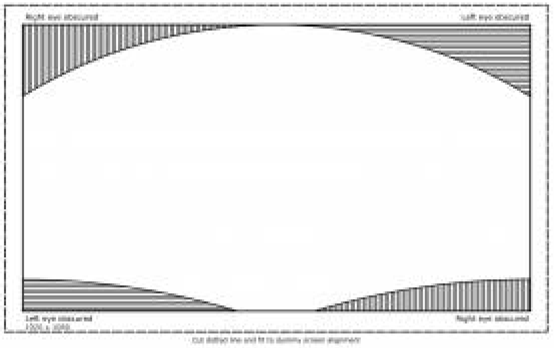
- A static pattern
- Eye tracking calibration dots
- Estimate of how corners are obstructed if viewed monocularly (*obstruction will vary with the individual*)

This dummy screen is used outside of the bore to help ensure a volunteer will be able see a suitable amount of the projected image and the eye tracking calibration dots when inside the bore.

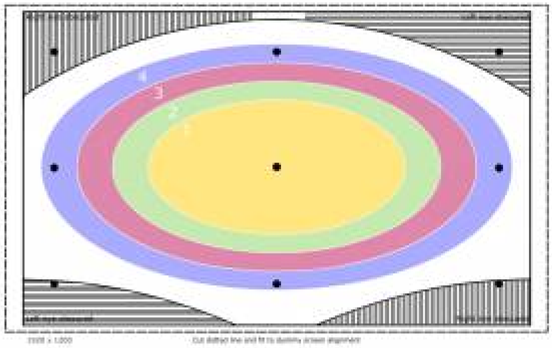
The below files can be used to customise a 'dummy screen' with different content.

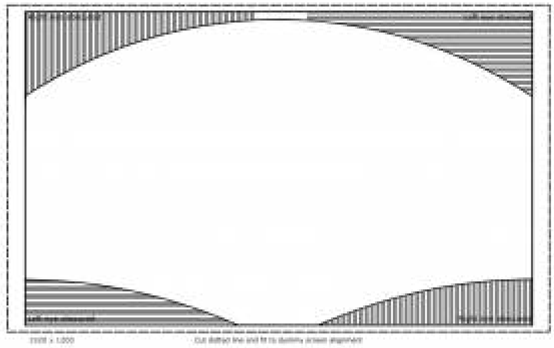
1920 x 1080





1920 x 1200





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